Can the models answer `test_prompts.txt`? Pretrained checkpoints and a negation fine-tune of approach_2

Experiment log, branch `attention-experiments`

2026-10-21

Summary

`test_prompts.txt` asks five questions whose answer is Paris (except P4), most of them preceded by a negated wrong city:

P1 The Eiffel Tower is in the city of
P2 The Eiffel Tower is not in Chicago. Therefore, the Eiffel Tower is in the city of
P3 (the sentence "The Eiffel Tower is not in Chicago." six times, then) Therefore, ... in the city of
P4 The Eiffel Tower is not in Paris. Therefore, the Eiffel Tower is in the city of
P5 The capital of France is not Berlin. Therefore, the capital of France is

- **No pretrained model answers them.** Every model that trained, including OpenAI's GPT-2 124M, answers P2 and P3 with the negated city (Chicago, 31–96% probability) and P4 with Paris. Repeating the negation (P3) makes them more confident in the wrong city, not less. They copy the city in the prompt rather than reason about "not". approach_2, approach_1, approach_3 and three_residual behave the same way as our GPT-2 baseline; approach_4, whose pretraining collapsed, emits commas, blank lines or "the the the".
- **Only OpenAI's GPT-2 knows the fact.** It is the only model with Paris as its top answer to P1 (7%). Our models, trained on 2.5B tokens, rank Paris 6th to 22nd.
- **A 400-step fine-tune removes the copying, but not the missing knowledge.** After fine-tuning on synthetic "X is not in D. Therefore, X is in the city of C" prompts (with Paris, France and the Eiffel Tower kept out), neither approach_2 nor GPT-2 + bias ever answers the negated city on held-out facts (from about 20%), and their accuracy with a negation equals their accuracy without one. On the test prompts they now answer P5 correctly (Paris: approach_2 44%, GPT-2 + bias 97%), but answer P1–P4 with London, Toronto or New York, because neither model knows the Eiffel Tower is in Paris.
- **approach_2 learns the skill as well as GPT-2; it just knows fewer facts.** Held-out accuracy after fine-tuning is 25–28% for approach_2 vs 33–36% for GPT-2 + bias, the same gap as before fine-tuning on plain questions, in line with approach_2's higher pretraining loss (3.627 vs 3.405).

1 Models and method

All checkpoints are the final 2.5B-token OpenWebText runs from the 2026-10-19 and 2026-10-20 reports (the self-logit bias is annealed to 0, so every model runs as its plain variant). approach_1 is its step-3000 checkpoint, saved before it diverged. "GPT-2 + bias" is standard GPT-2 124M trained with the same annealed bias, the control in those reports. OpenAI's pretrained GPT-2 124M is included as a reference: same architecture, far more training.

For each prompt we record the greedy continuation (8 tokens) and the next-token probability and rank (among all 50,257 tokens) of the correct city and of the city named in the prompt. Script: `train/ask_prompts.py`.

2 Pretrained models

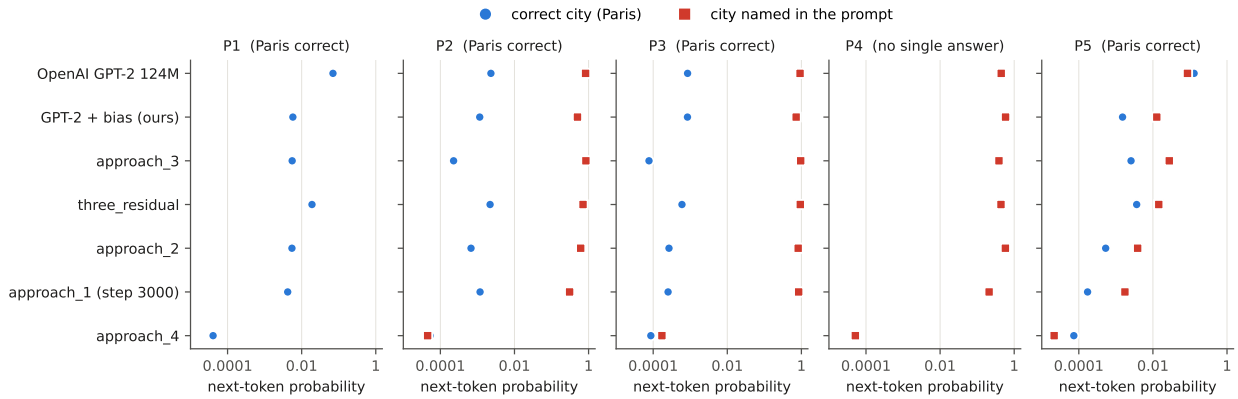


Figure 1: Next-token probability of Paris (blue circle) and of the city named in the prompt (red square: Chicago in P2–P3, Paris in P4, Berlin in P5). On P2 and P3 every working model puts 30–96% on Chicago and under 0.5% on Paris. Only OpenAI's GPT-2 gives Paris more weight than Berlin on P5.

model	P1	P2	P3	P4	P5
OpenAI GPT-2 124M	`Paris, and the...`	`Chicago.`	`Chicago. There...`	`Paris.`	`not Berlin. Th...`
GPT-2 + bias (ours)	`St. Paul, and ...`	`Chicago.`	`Chicago. The E...`	`Paris.`	`not Berlin.`
approach_3	`New York, and ...`	`Chicago.`	`Chicago.`	`Paris.`	`not Berlin.`
three_residual	`London, and th...`	`Chicago.`	`Chicago. The E...`	`Paris.`	`not the capita...`
approach_2	`San Francisco,...`	`Chicago.`	`Chicago. The E...`	`Paris.`	`not Berlin.`
approach_1 (step 3000)	`the city of th...`	`Chicago.`	`Chicago. The E...`	`Paris.`	`not Berlin. Th...`
approach_4		`,,,,`	`the the the th...`	`,,,,,,,,`	`the the the th...`

Table 1: Greedy continuation of each prompt (first line, cut at 14 characters).

	P1		P2		P3			P4	P5	
model	Paris	–	Paris	Chicago	Paris	Chicago	–	Paris	Paris	Berlin
OpenAI GPT-2 124M	0.070 (#1)	–	0.002 (#12)	0.831 (#1)	0.001 (#18)	0.921 (#1)	–	0.447 (#1)	0.129 (#2)	0.086 (#3)
GPT-2 + bias (ours)	0.006 (#15)	–	0.001 (#66)	0.501 (#1)	0.001 (#46)	0.723 (#1)	–	0.585 (#1)	0.002 (#64)	0.013 (#8)
approach_3	0.006 (#17)	–	0.000 (#115)	0.846 (#1)	0.000 (#64)	0.957 (#1)	–	0.388 (#1)	0.003 (#33)	0.028 (#5)
three_residual	0.019 (#6)	–	0.002 (#17)	0.709 (#1)	0.001 (#16)	0.935 (#1)	–	0.440 (#1)	0.004 (#26)	0.014 (#4)
approach_2	0.005 (#16)	–	0.001 (#60)	0.613 (#1)	0.000 (#66)	0.820 (#1)	–	0.579 (#1)	0.001 (#132)	0.004 (#21)
approach_1 (step 3000)	0.004 (#22)	–	0.001 (#75)	0.308 (#1)	0.000 (#82)	0.845 (#1)	–	0.210 (#1)	0.000 (#149)	0.002 (#20)
approach_4	0.000 (#2756)	–	0.000 (#2190)	0.000 (#2591)	0.000 (#1320)	0.000 (#594)	–	0.000 (#2254)	0.000 (#1468)	0.000 (#4792)

Table 2: Probability (rank) of the correct city and of the city named in the prompt.

Three observations:

1. **Copying wins over negation.** In P2–P4 the most likely answer is always the city in the prompt. This is the known behavior of small language models ("not" barely changes the next-token distribution), and the architecture changes do not affect it: all working variants, ours and OpenAI's, agree on every greedy answer for P2–P4.
2. **Repetition strengthens it.** Going from one negated sentence (P2) to six (P3) raises P(Chicago) for every working model, e.g. 0.61 → 0.82 for approach_2.
3. **The fact itself is weak.** On P1 our models put 0.5–2% on Paris; three_residual ranks it 6th, the others 15th–22nd. OpenAI's GPT-2 ranks it 1st (7%). P5 is similar: only OpenAI's GPT-2 prefers Paris over Berlin.

3 Fine-tuning approach_2 to ignore negated cities

3.1 Setup

The question is whether approach_2 can be taught to ignore a negated distractor, and whether that skill transfers to facts it was not fine-tuned on. GPT-2 + bias gets the identical fine-tune as a control.

Data. Hand-written facts: 45+14 landmarks (landmark → city) and 69+22 countries (country → capital). Every fact involving Paris, France or the Eiffel Tower is excluded, so the five test prompts stay unseen. A random 25% of each list is held out for evaluation:

- held-out landmarks: The Golden Gate Bridge, The Acropolis, The Sydney Opera House, Camp Nou, The Bund, The Forbidden City, The CN Tower, Buckingham Palace, The Statue of Liberty, The Rijksmuseum, The Petronas Towers, Anfield, The Gateway Arch, The Grand Ole Opry.
- held-out countries: the Philippines, Rwanda, Morocco, Uruguay, Jamaica, Zimbabwe, Serbia, Hungary, Japan, Sweden, Azerbaijan, Malaysia, Italy, Iraq, Venezuela, Sudan, Bangladesh, Chile, Ghana, Jordan, Canada, Croatia.

Each training example uses one of three forms, sampled 1:2:1:

- plain: "The Colosseum is in the city of" → " Rome."
- negated: "The Colosseum is not in Oslo. Therefore, the Colosseum is in the city of" → " Rome."
- repeated: the negated sentence 2–6 times, then the question.

Countries use "The capital of X is (not D). Therefore, the capital of X is". The distractor D is a random other city from the pool of answer cities plus 18 large cities (Chicago among them).

Training. 400 steps, AdamW ($\beta = (0.9, 0.95)$, no weight decay), learning rate 3×10^{-5} after a 20-step warmup, gradient clip 1.0. Each step has 64 synthetic prompts, with the loss on the answer tokens only, plus 8 OpenWebText sequences of 1024 tokens with the normal language-model loss, to limit forgetting. One run per model, seed 0, about 1.5 minutes on one GPU. Script: `train/finetune_negation.py`.

Evaluation. For each entity and form, greedy-decode as many tokens as the answer has. A prompt is *correct* if the decoded text equals the city, and counts as *answering the negated city* if it starts with the distractor. OpenWebText loss is measured on the first 64 validation sequences (65k tokens), so it is comparable before vs after but not to the full-split numbers in earlier reports.

3.2 Results

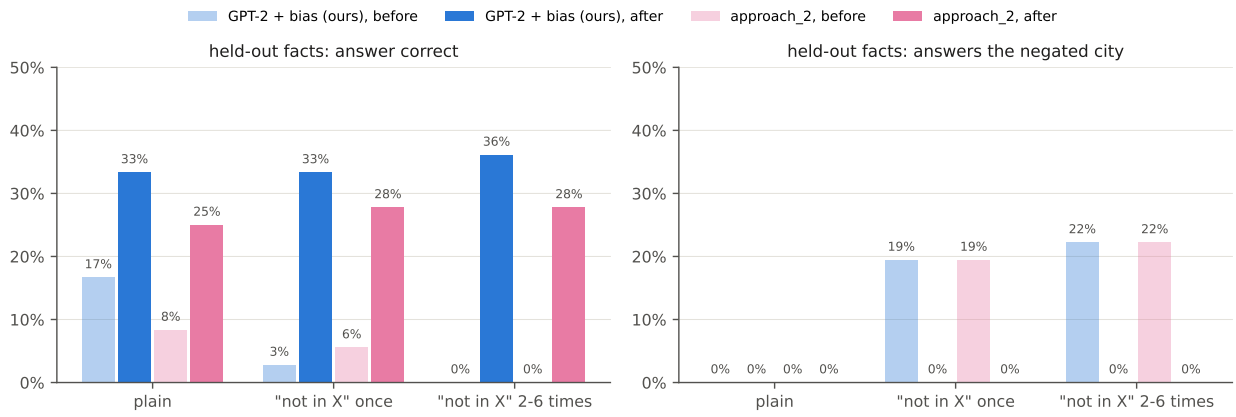


Figure 2: Held-out facts (36 entities, so one prompt is 2.8 points), before (faded) and after (solid) fine-tuning. Left: accuracy. Right: how often the model answers with the negated city.

	held-out: correct			held-out: negated city		train ents.	OWT
model	plain	once	2–6×	once	2–6×	once, correct	loss
approach_2, before	8%	6%	0%	19%	22%	0%	3.675
approach_2, after	25%	28%	28%	0%	0%	100%	3.731
GPT-2 + bias (ours), before	17%	3%	0%	19%	22%	0%	3.459
GPT-2 + bias (ours), after	33%	33%	36%	0%	0%	100%	3.499

Table 3: Fine-tuning metrics. "once" and "2–6×" are the negated forms.

- **The skill transfers to unseen facts.** Before fine-tuning, both models answered the negated city on about 20% of held-out negated prompts and were almost never right (0–6%). Afterwards, they never answer the negated city, and accuracy is the same with or without the negation (approach_2: 25% plain, 28% negated; GPT-2 + bias: 33% vs 33–36%). The fine-tune taught "ignore the negated city", not just answers.

- **Plain-fact accuracy also rose** (approach_2 8% → 25%, GPT-2 + bias 17% → 33%). The question format and the answer format (a city, then a period) became familiar, so the models now produce knowledge they already had but did not express.
- **Knowledge sets the ceiling, and approach_2 has less.** approach_2 trails GPT-2 + bias by about 8 points on every form, both before and after, consistent with its higher pretraining loss.
- **Cost.** OpenWebText loss rose by 0.056 for approach_2 and 0.040 for GPT-2 + bias. The training-entity answers were memorized (answer loss below 10^{-3}, 100% correct), see Figure 4.

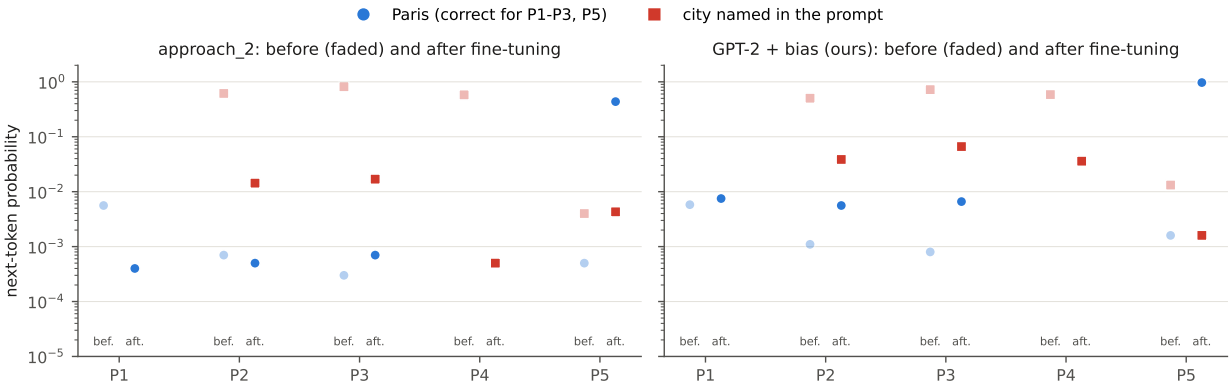


Figure 3: Test prompts before (faded) and after fine-tuning. Red squares: the city named in the prompt; blue circles: Paris.

model	P1	P2	P3	P4	P5
approach_2, before	`San Francisco,...`	`Chicago.`	`Chicago. The E...`	`Paris.`	`not Berlin.`
approach_2, after	`Toronto.`	`Toronto.`	`London.`	`London.`	`Paris.`
GPT-2 + bias (ours), before	`St. Paul, and ...`	`Chicago.`	`Chicago. The E...`	`Paris.`	`not Berlin.`
GPT-2 + bias (ours), after	`New York.`	`New York.`	`New York. The ...`	`New York.`	`Paris.`

Table 4: Greedy answers to the test prompts before and after fine-tuning.

On the test prompts (Figure 3, Table 4):

- **The copying is gone.** P(Chicago) on P2–P3 fell from 0.5–0.8 to 0.01–0.07, and P4 no longer answers Paris.
- **P5 is solved** by both models (Paris: approach_2 0.44, GPT-2 + bias 0.97), since both know France's capital once the Berlin distractor is ignored.
- **P1–P3 are not.** Both models now answer with a frequent training answer (approach_2: Toronto or London; GPT-2 + bias: New York). For approach_2, P(Paris) on P1 even dropped (0.006 → 0.0004).

The drop has a likely cause in the data design. Paris was never an answer during fine-tuning, while London and New York were the most frequent answers (4 training entities each), so the fine-tune shifts probability toward those cities. For a model that does not know the fact, the frequent answers

win. GPT-2 + bias's P(Paris) on P1–P3 rose slightly (0.006–0.007 after), but stayed far below its top answer.

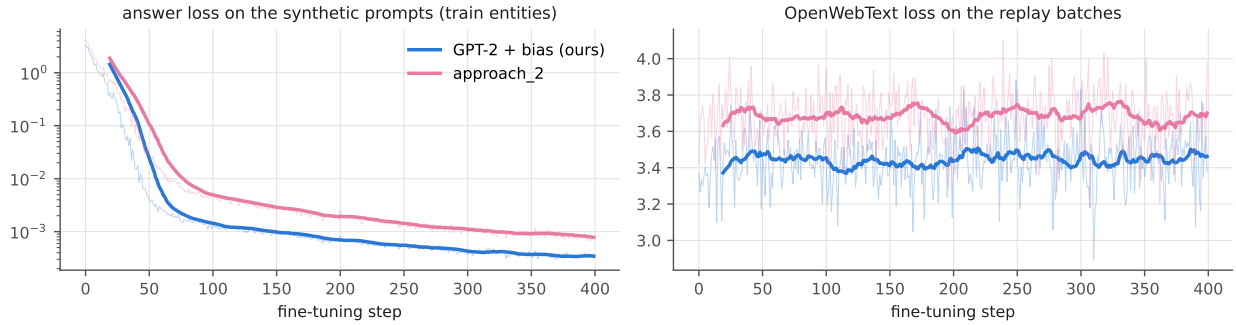


Figure 4: Fine-tuning curves (thin: per step; thick: 20-step mean). Left: answer loss on the synthetic training prompts, log scale. Right: loss on the OpenWebText replay batches, which stays at each model's pretraining level.

4 Limitations

- One seed per configuration, and only 36 held-out entities (108 prompts), so each held-out percentage is uncertain by roughly ±8 points. The approach_2 vs GPT-2 + bias gap after fine-tuning is about one such interval.
- The facts list is small and hand-written, and the templates are fixed. The fine-tune likely teaches this exact phrasing; other forms of negation ("isn't", "rather than") were not tested.
- Entities are held out, but answer cities are not: a held-out landmark in London has an answer the fine-tune trained on, which can inflate held-out accuracy. The negated-city rate (20% → 0%) does not depend on this.
- Excluding all Paris answers protects the test prompts but biases the models away from Paris (above).
- P4 has no single correct answer, so it is only used to check that the model stops repeating the city in the prompt.

5 Possible next steps

- Add plain "The Eiffel Tower is in the city of Paris." (and other Paris facts) to fine-tuning, but no negated Eiffel prompts. P2–P4 would then test only the negation skill, with knowledge out of the way.
- Mix several negation phrasings and test on unseen ones, to see whether the skill is about "not" or about this template.
- Fine-tune OpenAI's GPT-2 124M the same way: it already knows the Eiffel Tower fact, so it shows what the fine-tune achieves when knowledge is not the limit.
- Repeat with 3 seeds and a larger held-out set to resolve the approach_2 vs GPT-2 + bias gap.

Files

- `data/`: `prompts_pretrained.json` (all pretrained models), `ft_approach_2_results.json` and `ft_baseline_results.json` (before/after metrics, entity split, test-prompt outputs), `ft_*_log.jsonl` (per-step losses), `test_prompts.txt`.
- `code/`: `ask_prompts.py`, `finetune_negation.py`, `make_negation_report.py`.
- Fine-tuned weights: `/local/scratch/a/liu3170/runs/openwebtext/ft_negation/approach_2/finetuned.pt` and the same path with `baseline`.